A positive definite counterexample to the nonlinear quantum-measurement monotone

Complete solution of MathDB 359456

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Task 1. Resolve the nonlinear monotone conjecture. A blockwise probability vector $\boldsymbol{P} = (P_1, \dots, P_n)^{\mathsf{T}}$ consists of positive semidefinite $d \times d$ matrices summing to I_d. A blockwise bistochastic matrix $\boldsymbol{B} = (B_{ij})$ has positive semidefinite blocks, with every block row and block column summing to I_d. The source's sequential product is

$$Q_i = (\boldsymbol{B} * \boldsymbol{P})_i = \sum_{j=1}^{n} P_j^{1/2} B_{ij} P_j^{1/2}.$$

Conjecture 1 of Rico and Życzkowski asks whether, for every ordering of the Q_i, some ordering of the P_i makes

$$\left\| \sum_{i=1}^{k} (P_i - I_d/n) \right\|_2 \geq \left\| \sum_{i=1}^{k} (Q_i - I_d/n) \right\|_2 \quad (1 \leq k \leq n), \qquad \|X\|_2^2 = \operatorname{tr}(X^* X).$$

The answer is negative, already for $n = d = 3$ and diagonal positive definite effects and blocks.

Let

$$P_1 = \frac{1}{12} \operatorname{diag}(10, 1, 1), \quad P_2 = \frac{1}{12} \operatorname{diag}(1, 10, 1), \quad P_3 = \frac{1}{12} \operatorname{diag}(1, 1, 10),$$

and take

$$\boldsymbol{B} = \begin{pmatrix} P_1 & P_2 & P_3 \\ P_2 & P_3 & P_1 \\ P_3 & P_1 & P_2 \end{pmatrix}.$$

Each P_j is positive definite, and $P_1 + P_2 + P_3 = I_3$. Every block row and column of $\boldsymbol{B}$ is a permutation of this triple. Thus all hypotheses hold, with all nine blocks positive definite as well.

All matrices involved are diagonal, so $P_j^{1/2} B_{ij} P_j^{1/2} = P_j B_{ij}$. Direct multiplication yields

$$Q_1 = P_1^2 + P_2^2 + P_3^2 = \frac{17}{24} I_3, \qquad Q_2 = Q_3 = P_1 P_2 + P_2 P_3 + P_3 P_1 = \frac{7}{48} I_3.$$

These output effects are also positive definite and sum to I_3.

Consider the displayed output ordering, with Q_1 first. For every $j = 1, 2, 3$, the diagonal entries of $P_j - I_3/3$ are a permutation of $(1/2, -1/4, -1/4)$, whence

$$\|P_j - I_3/3\|_2^2 = \frac{1}{4} + \frac{1}{16} + \frac{1}{16} = \frac{3}{8} = \frac{24}{64}.$$

In contrast,

$$\|Q_1 - I_3/3\|_2^2 = \left\|\frac{3}{8}I_3\right\|_2^2 = \frac{27}{64} > \frac{24}{64}.$$

No choice of the first input effect can satisfy the required inequality at $k = 1$. Therefore no input ordering works for this output ordering, which disproves the conjecture. □

Scope and verification. The counterexample uses three effects and lies outside the source's restricted class of sortable measurements: for instance, $P_1 - P_2 = \mathrm{diag}(3/4, -3/4, 0)$ is indefinite. The conjecture explicitly concerns measurements beyond that class. All effects and blocks are positive definite, so the failure does not depend on zero effects or singularities. The accompanying script checks the sequential product, all normalizations, positivity and the squared-norm gap 3/64 exactly. All thirteen checks pass, including all six input orderings.

Task 2. Quantify the possible increase. Bound the increase of $\max_i \|P_i - I_d/n\|_2$ under blockwise bistochastic dynamics, in terms of the number of effects and the dimension.

Task 3. Find valid nonlinear monotones. Identify nonlinear quantities that are monotone on all quantum measurements under the source's sequential product.

Task 4. Characterize the diagonal case. For simultaneously diagonal effects and blocks, determine when a single ordering of the input effects controls every required partial sum for a specified output ordering.

Task 5. Determine conditions restoring the conjecture. Find structural conditions beyond the source's sortable class that ensure the asserted inequalities, and determine whether they are preserved by blockwise bistochastic dynamics.

References

[1] A. Rico and K. Życzkowski, *Discrete dynamics in the set of quantum measurements*, Journal of Physics A: Mathematical and Theoretical **57** (2024), 435302. Appendix C.1, Conjecture 1, equation (C.1); published article, arXiv:2308.05835, version 2, p. 28.